

## Boxplot Example

Brookman Electronics manufactures a new type of electronic component and is examining the lifetime of the components. Engineers are studying two different manufacturing processes named A and B. Engineers are concerned that Process B has about a 10% defective rate: about one out of every ten components made with Process B has a lifetime of less than two hours. Last year, one-hundred components were produced with each process. Engineers recorded the lifetime of each component and rounded off to the nearest hour. The ranked data is on the last page.

For each data set find Q1, median, Q3, IQR, inner fences, and outer fences. Then draw side-by-side boxplots.

|           | A | B |
|-----------|---|---|
| n         |   |   |
| $I_m$     |   |   |
| median    |   |   |
| $I_q$     |   |   |
| $Q_1$     |   |   |
| $n+1-I_q$ |   |   |
| $Q_3$     |   |   |
| IQR       |   |   |
| step      |   |   |
| UIF       |   |   |
| LIF       |   |   |
| UOF       |   |   |
| LOF       |   |   |

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|                | A                                            | B <i>→ (do it by yourself)</i> |
|----------------|----------------------------------------------|--------------------------------|
| n              | 100                                          | 100                            |
| $I_m$          | $I_m = \frac{n+1}{2} = \frac{101}{2} = 50.5$ | 50.5                           |
| median         | $\frac{509 + 509}{2} = 509$                  |                                |
| $I_q$          | $\frac{n+2}{4} = \frac{102}{4} = 25.5$       | 25.5                           |
| Q <sub>1</sub> | $\frac{445 + 447}{2} = 446$                  |                                |
| $n+1-I_q$      | $100+1-25.5 = 75.5$                          |                                |
| Q <sub>3</sub> | $\frac{565 + 566}{2} = 565.5$                |                                |
| IQR            | $Q_3 - Q_1 = 119.5$                          |                                |
| step           | $1.5 (IQR) = 179.25$                         |                                |
| UIF            | $Q_3 + \text{step} = 744.75$                 |                                |
| LIF            | $Q_1 - \text{step} = 266.75$                 |                                |
| UOF            | $Q_3 + 2\text{step} = 924$                   |                                |
| LOF            | $Q_1 - 2\text{step} = 87.5$                  |                                |

$$I_q = \begin{cases} \frac{n+2}{4} & n \text{ is even} \\ \frac{n+3}{4} & n \text{ is odd} \end{cases}$$

haven't been shown in the boxplot below

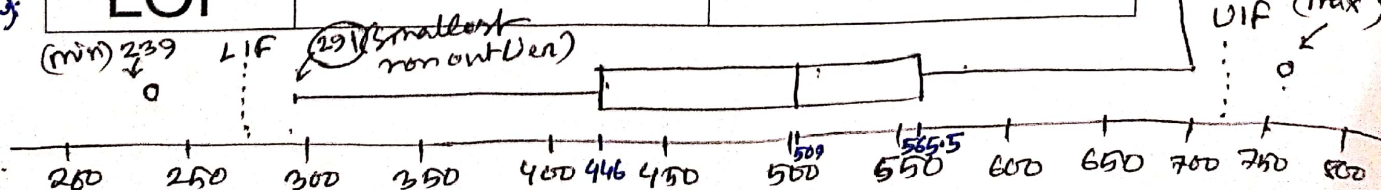
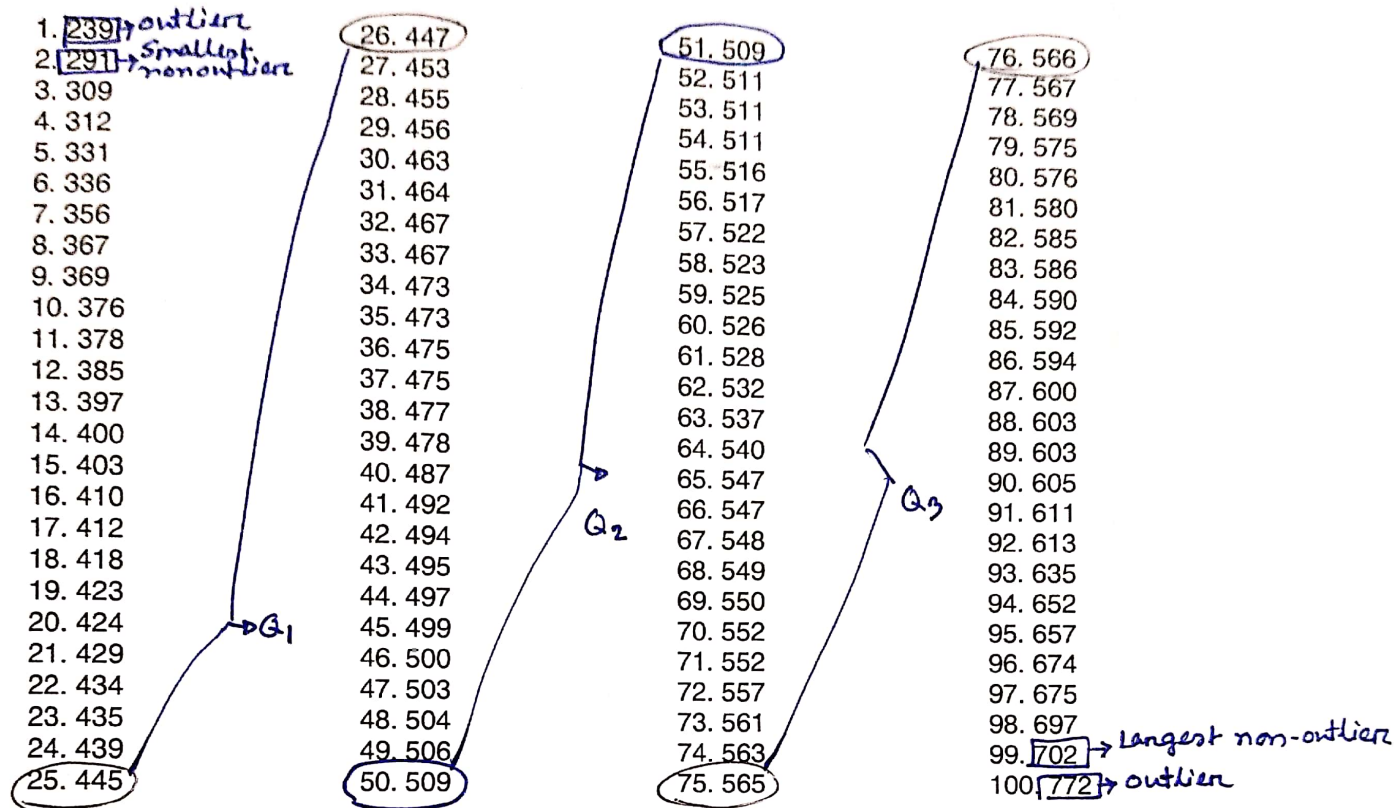


Fig: Box plot of A.

# Process A:



## Process B:

|         |                |                |          |
|---------|----------------|----------------|----------|
| 1. 0    | 26. 414        | <u>51. 491</u> | 76. 551  |
| 2. 0    | 27. 420        | 52. 499        | 77. 554  |
| 3. 0    | 28. 423        | 53. 501        | 78. 560  |
| 4. 0    | 29. 425        | 54. 502        | 79. 560  |
| 5. 0    | 30. 425        | 55. 505        | 80. 563  |
| 6. 1    | 31. 426        | 56. 506        | 81. 565  |
| 7. 1    | 32. 427        | 57. 507        | 82. 567  |
| 8. 1    | 33. 437        | 58. 508        | 83. 580  |
| 9. 1    | 34. 438        | 59. 509        | 84. 580  |
| 10. 1   | 35. 439        | 60. 514        | 85. 581  |
| 11. 221 | 36. 443        | 61. 515        | 86. 582  |
| 12. 321 | 37. 447        | 62. 519        | 87. 587  |
| 13. 325 | 38. 457        | 63. 523        | 88. 587  |
| 14. 339 | 39. 458        | 64. 523        | 89. 598  |
| 15. 348 | 40. 460        | 65. 524        | 90. 601  |
| 16. 352 | 41. 462        | 66. 528        | 91. 613  |
| 17. 390 | 42. 462        | 67. 528        | 92. 634  |
| 18. 392 | 43. 465        | 68. 529        | 93. 641  |
| 19. 394 | 44. 468        | 69. 533        | 94. 644  |
| 20. 400 | 45. 476        | 70. 534        | 95. 648  |
| 21. 403 | 46. 477        | 71. 534        | 96. 659  |
| 22. 408 | 47. 484        | 72. 535        | 97. 663  |
| 23. 408 | 48. 485        | 73. 537        | 98. 665  |
| 24. 410 | 49. 486        | 74. 543        | 99. 691  |
| 25. 414 | <u>50. 487</u> | 75. 549        | 100. 698 |

Process A and B:  $n = \boxed{100}$

Process A and B:  $l_m = \frac{n+1}{2} = \frac{101}{2} = \boxed{50.5}$

Process A and B:  $l_q = \frac{n+2}{4} = \frac{102}{4} = \boxed{25.5}$

Process A and B:  $n+1-l_q = 100+1-25.5 = \boxed{75.5}$

$$\text{Process A median: } \frac{509 + 509}{2} = \boxed{509}$$

$$\text{Process A } Q_1: \frac{445 + 447}{2} = \boxed{446}$$

$$\text{Process A } Q_3: \frac{565 + 566}{2} = \boxed{565.5}$$

$$\text{Process A IQR: } 565.5 - 446 = \boxed{119.5}$$

$$\text{Process A step: } 1.5(\text{IQR}) = 1.5(119.5) = \boxed{179.25}$$

$$\text{Process A UIF: } 565.5 + 179.25 = \boxed{744.75}$$

$$\text{Process A LIF: } 446 - 179.25 = \boxed{266.75}$$

$$\text{Process A UOF: } 565.5 + 2(179.25) = \boxed{924}$$

$$\text{Process A LOF: } 446 - 2(179.25) = \boxed{87.5}$$

$$\text{Process B median: } \frac{487 + 491}{2} = \boxed{489}$$

$$\text{Process B } Q_1: \frac{414 + 414}{2} = \boxed{414}$$

$$\text{Process B } Q_3: \frac{549 + 551}{2} = \boxed{550}$$

$$\text{Process B IQR: } 550 - 414 = \boxed{136}$$

$$\text{Process B step: } 1.5(\text{IQR}) = 1.5(136) = \boxed{204}$$

$$\text{Process B UIF: } 550 + 204 = \boxed{754}$$

$$\text{Process B LIF: } 414 - 204 = \boxed{210}$$

$$\text{Process B UOF: } 550 + 2(204) = \boxed{958}$$

$$\text{Process B LOF: } 414 - 2(204) = \boxed{6}$$